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Section A : Data Fundamentals

1. A database is a structured system for storing, managing, and retrieving data, typically stored electronically in a computer system.
- database are essential in the moden world because they store large amount of data or information efficiently without the system being slow.
- It also allows multiple users to have access to the same file, be able to read and write information at the exact same time without crashing.
- Banking systems use database to store and track customer information, transaction, and account balances.
- Online shopping platforms manages customer details, orders, products and inventory.

2. A Data Warehouse is a specialized type of data managemne system designed specifically to specifically to support business intelligence and data analytics.
- A data warehouse is built and act as a central repository where data from many different sources like sales, marketing, and inventory is consolidated for high-level decision making.
- A company would use a data warehouse because data warehouse are optimized to run massive analytical query allowing analysts to run complex query/queries without affecting live business operations.

3) Data Lake → is a centralized, highly flexible repository that allows you to store all your structure and unstructured data at any scale.

→ Unlike a data warehouse which requires data to be cleaned and formatted before it can be saved, a data lake allows you to store data in its raw, native format, including Photos, videos, Pdfs and sensor logs without having to structure it first.

→ A company would prefer a data lake over a data warehouse if a company needs to store data ~~that~~ that does not fit neatly into rows and column, such as social media posts, images or audio files from customer service calls.

4. A Data Mart is a focused, specialized version of a data warehouse that contains a subset of data tailored to a specific department or business line. If a data warehouse is a massive central library containing every book in the city, then a data mart is the science or history section. It holds only what a specific group needs to do their job.

→ Data mart are built for business level users rather than data scientist.

(5) Structure data is highly organized and follows a rigid, predefined format or schema. (e.g a bank statement.

- Semi-structured data → doesnot reside in a fixed grid like a table, but it still contains internal tags or markers that seperate data elements and enables hierarchies. E.g Email.

- Unstructured Data → is a data that has no predefined format or organisation. It's usually heavy data that is qualitative in nature and cannot be easily process by traditional tools. e.g a video file.

(6) Data Modeling → is the process of creating a visual representation or blueprint of an entire information system or part of it.
It defines the specific data used and the relationship between different data points.

* Conceptual data model → it focuses on entities like product or customers and their relationships.
* logical Data Model → This model adds a layer of technical detail to the conceptual model. It defines the specific attributes for each entity and establishes the primary keys that links tables together.

(7) Data Mining → is the process of sorting through massive dataset to identify patterns, trends and anomalies that aren't immediately obvious.
- Business use data mining to move from reactive decision making by looking at what happened to proactive decision-making by predicting what will happen.

E.g Retailer use data mining to find associations between product. by analyzing millions of checkout receipts, they might discover that people who buy meet on friday evenings are also likely to buy alcohol and ice.

(8) A Database Managent System (DBMS) is the software that interacts with end users, applications, and the database itself to capture and analyse data.

* MySQL → it's the most popular open-source relational database. owned by Oracle. It is known for being reliable to setup, and supported by almost every web hosting provider.
→ The best use case is websites and E-commerce. It's perfect for applications that need high-speed read operations and proven track record.

* PostgreSQL → is an advanced, Open-Source relational database it is favoured Mostly by developers who need more complex features than MySQL offers, Such as Support for custom data types and Extremely Complex queries.

* The Suitable use for it, is Data analysis and complex systems Where people or user want to use the Same data at once

* Oracle database → is a high-end, Commercial relational database design for large-scale Enterprise environments. It is packed with advanced Security, backup, and recovery features, but it comes with significant licensing cost.

→ The best use is in banking and Large Enterprise Infrastructure.

* (8) MongoDB → is the leading NOSQL database that uses flexible, JSON-like documents to Store data, instead of using tables with Rows and Columns.

* Best use Case is Real-time management and big data, It is Ideal for Social media feeds, mobile apps and or any other platform where the data structure changes frequently.

(9) ETL (Ettract, Transform, Load) is a three-step data intergration process used to collect data from various sources, alter it into a usable format, and store it in a Central repository.

The three Steps.

(a) Extract.

The first Step is to gather data from multiple different Sources. the goal is to get all the raw data into a Single Staging area without disturbing the Original Source.

9.2 Transform.

- This is the most critical stage. Raw data is often messy, inconsistent or formatted differently depending on where it came from.
During ~~data~~ transformation, the data is cleaned and mapped.
- This where data errors are identified and corrected.

9.3 Load

This is the final stage/step, where the newly cleaned and formatted data is written into its final destination typically a data warehouse.

This process is important because it ensure that all the data is found one source, and it is clean, consistent and reliable. This allows a data warehouse to be a single source of the truth where all the data is kept.

10.1) CSV - (Comma-Seperated Values)

- This is the most comonen or basic and universal data format. It stores data as plain text, where each line is recorded and each value is seperated by a comma.
I would use CSV for small dataset or when there is a need to share data with non-technical user who use Excel instead of JSON.

10.2 * JSON (Java Script Object Notation)

- is a text based format that uses key-value pairs. It is the standard for web applications and APIs.
Use JSON when you are exchanging data between web services or when your data structure is unpredictable and changes frequently.

10.3 Parquet

Parquet is a Columnar Storage format. It is unlike CSV, which stores data row-by-row, Parquet stores it column-by-column. It is highly Compressed and optimized for big data.

- You can use Parquet for large-scale analytics and data warehousing

10.4 Avro

- Avro is a row based format that is used Extremely in Streaming data. It stores the Schema along with the data itself in a compact binary format. It is design to be fast for heavy writing operations where data is constantly being added.
- Use Avro for real-time data pipelines.
- It is excellent when you need to move data between systems quickly and ensure that the data structure never breaks.

Section B: Artifial Intelligence and Machine learning.

11. Artificial Intelligence (AI) is a field of Computer Science dedicated to Creating Systems capable of performing tasks that traditionally requires human intelligence. These tasks includes things like patterns recognition, desision making, translating languages and learning from experience.

Narrow AI vs General AI

Narrow AI

+ Narrow AI is designed to perform a single, specific task or a limited range of tasks. It operates under a predefined set of constraints and doesn't possess consciousness or intelligence outside of its speciality.

Example: Recommendation systems, Facial Recognition Systems

General AI

is a theorectical form of AI that would possess the ability to understand, learn, and apply its intelligence to any intellectual task that a human being can. General AI would think and understand, not just execute tasks.

12. Machine Learning. (ML)

- Machine learning is a subfield of Artificial Intelligence that focuses on building systems that can learn from data and improve their performance over time without being ~~the~~ explicitly programmed for every specific scenario.
- While traditional progamming relies on human to write down every rule/code the computer must follow, ML allows the computer to discover those rules itself by identifying patterns in large dataset.

Three types of Machine learning.

12.1 Supervised Learning

12.2 Un Supervised learning

12.3 Reinforcement Learning.

12:1 Supervised Learning

Also called the "Teacher" Approach — in this type, the model is trained on a labeled dataset. This means for every piece of data you give the computer you also give it the correct answer. In this way you train it to be right all the time or when it's wrong, so that it learns the correct information.

12.2 Un Supervised Learning.

→ This one is also called the Pattern Finder. Here; the data has no labels and no correct answers. The computer is simply told to look at the data and find any interesting structures or groupings on its own. It uses mathematical techniques to find similarities or anomalies that a human might miss.

12.3 Reinforcement Learning.

— Also known as The Trial and Error Approach. This is inspired by how how humans and animals learn. An agent (the AI) is placed in an Environment and given the goal. It receives rewards for good action and penalties for mistakes. It does not learn from a static dataset; It learn by interacting with world.

13. Deep Learning (DL)

→ Deep learning is a specialized subfield of machine learning that uses multi-layered Artificial Neural Networks to solve complex problems.

It is the technology behind the most humanlike AI we see today, such as voice assistanes, self-driving cars, and facial recognition.

Neural Networks.

An Artificial Neural network is a computational model inspired by the structure of the human brain. The similar way our brain are made of interconnected neurons that sends signals to one another, an Artificial Neural Network is made of layers of nodes.

How Neural network works.

A neural Network typically consist of Three main layers.

(i) Input layer → This is where the data enters the system, Each node represents a Single feature of that data.

(ii) Hidden Layers → This is where the deep in deep learning comes from. There can be dozens or even hundreds of these layers. Each layer take the information from the Previous one and extract more complex feature.

(iii) Output layer → the Final layer that Provides the results or Predition

14.) Natural Language Processing (NLP)

→ Natural Language Process (NLP) is a branch of Artificial Intelligence that focuses on enabling Computers to understand, interprete, and generate human Language. (text/speech).

- It allows machines to communicate with humans in natural language, like English, or any other language.

Examples.

(i) When you Ask your phone "What's the best restaurant near me", NLP is working behind the Scenes. (virtual Assistants)

(ii) When you are typing on Google, or email, your phone/computer is Constantly predicting the next word you want to use. (Predictive Text and Autocorrect.

(iii) Services like Gmail and Outlook use NLP to keep your inbox Organised by Filtering your Emails and Sorting. It scans the language of an incoming mail to spot susficious patterns or common scam phases.

15.) Large Language Model (LLM)

A Large Language Model is a type of Artificial Intelligence trained on massive amount of text data to understand, Summarize, and generate human-like language.

The term "Large" in its name refers to "massive dataset" and "huge parameter count".

* LLMs are powerful, flexible systems designed to work with human language at scale.
* They go beyond traditional ML by being general-purpose and conversational.

Two well known LLMs

(i.) GPT (Generative Pre-trained Transformer) → They help you study, write letters, or debug code.

(ii.) BERT (Bidirectional Encoder Representations from Transformers) → They improve search engines understanding, text classification and sentiment analysis.

16. Generative AI

Is a type of AI that can create new content, such as text, images, audio, video, and even computer codes, rather than just analysing or classifying existing data.

* At its high level-Generative AI works by learning the underlying patterns, and structures of the data it was trained on. Once it understands those pattern, it can generate new data that has similar characteristics.

Three tools that use Generative AI

(i) ChatGPT → It answers questions, write essays and helps with coding.

(ii) Gitthub → helps developers write code faster and reduce errors.

(17.) Training datasets

* Is the initial set of data use to teach a machine learning model how to make predictions or perform a specific task.
* The quality of training data is critical because AI models donot have a common sense or external knowledge it is only dependant on what you're feeding it or teaching it.

Incorrect data or bias.

When the training data is flawed, the AI model inherits those flaws. This leads to serious issues such as Algorithmic bias, Incorrect logic, Memorization Trap and poor safety and security.

18. Computer vision

- Computer Vision is a field within Artificial Intelligence that enables machines to interpret and understand visual information from the world, such as images and videos.

* Machines don't actually see like humans do; they process images as numerical data. An images is converted into a grid of pixels each pixels has a numerical value (e.g., RGB values for colour)

Using models from deep learning, early layers detect simple features (edges, lines)

Deeper layers detect complex pattern. like (Shapes and objects)

Real world Examples.

(i) Facial recognition is used in smartphones and security systems.

(ii) X-ray images for detecting diseases

Section C: APIs, Tools and Developer Concepts.

19 AN API (Application, Programming) is a set of rules and protocols that allows different Software applications to communicate with each other. It act as an intermediary, taking a request from one system, delivering it to another, and then bringing the respons back.

Analogy:

* you = the user
* Remote control (API) = the interface.
* TV (system/software = where the actual work happens.

How it works

(i) you press a button on the remote. (request)

(ii) The remote send the signal to the TV

(iii) The TV process the signal (change the channel/volume)

(iv) you see the results instantly (response.)

Importance of APIs

For data professionals APIs are the primary way to automate data collection. Instead of manually downloading a csv every morning, a script can use an API to automatically pull the latest results into a data warehouse for analysis.

This means APIs allows to acces data from a server without being physically there or using hard drives.

20. An API key - is a unique string of characters (letter, number and symbols) that a software service providers to a developer. When an application makes a request to a server, it "attaches" this key to prove its identity.

* API keys serves as the first line of defense in protecting valuable digital resources and services. This ensures only authorized users/apps can access the API.

Best Practice for Storing and protecting API keys.

* It must never be hard-code keys in source code
  * Store them in secure configuration files, Environment variables, and secret management tools.

* If you accidentally Share your API key in Public sites and bots will usually find it whin seconds. The Consequences can be severe: hackers can have acces into private information, such bank information and steal your money.
Attackers Could use the key to steal sensitive Customer information or delete your entire data base.

21. REST API

* A REST API is an Application programming interface that follow the design Principle of the REST architectural style.
* REST - is Short to Representational State Transfer, and is a set of rules and guidelines about how you should build a web API.
* Main HTTP Methods in REST APIs

(i) GET → Used to retrieve data from a Server and does not modify anything.

(ii) POST → Used to send new data to the Server to create a new resource. Unlike "GET", the data is sent in the body of the request rather than the URL.

(iii) PUT → The "PUT" method is used to update an existing resource or create it if it doesn't exist. Crucially "PUT" usually replaces the entire resource with new data provided.

(iv) DELETE → The "Delete" method is used to remove a Specific resource from the Server.

2.2. JavaScript Object Notation (JSON)

* It is a lightweight, text-based format used for storing and transporting data. Although it originated from JavaScript, it is language-independant, meaning almost every modern programming language can read and write JSON easily.

* JSON vs API and Data Exchange

JSON replaced older formats like XML because it is specifically optimized for the Morden web:

- It uses a simple key-value structure that is easy for a person to look at and understand immediately.
- It is machine friendly - Incredibly faster for computers to parse. Which works best for mobile devices and servers.

Json object Example.

"name": Siyabonga → key = name, value = Siyabonga

"skills": [...] a list (array) of items.

"Address": {...} → a nested object.

2.3 Prompt Engineering.

→ is the process of intentionally designing, refining, and optimizing inputs (prompt) to guide larger Large Language Models (LLM) toward producing the most accurate, high quality, and relevant outputs.

If an AI model is a high performing engine, prompt engineering is the skill of driving it precisely. It involves understanding the model's capabilities and limitations to nudge it into the right persona, tone, and logical framework.

Prompt Engineering is important because:

* Reduce Hellucinations → Clear prompts gives the AI a narrow sandbox to play in.
* Saves Time → It gets results on the first try.

Poor Prompt

"Write a blog post about Gas Chromatography"

Improved Prompt.

"Write a 500-word educational blog post about the basics of Gas Chromatography. The target audience is entry-level lab technicians. Use a professional yet encouraging tone. Include a section "Common troubleshooting Tips" and format the post with markdown headers."

24 MAP function

* The MAP function is a higher-order function used to transform every item in a collection (like a list or an array) into something else, without changing the original collection. It transforms data by applying the same rule to each element.

* Input list → Apply function → Output list.

Real world example.

Goal → Convert temperatures fro Celsius to Kelvin.

Python

```
Celsius = [0,10,20,30]
Kelvin = list(map(lambda c: (c* 9/5 + 32, Celsius)) Print(Kelvin)
```

25. Cloud Computing.

* Is the on demand delivery of computing services, including servers, storage, database, networking and software, over the internet (the cloud)

* Instead of owning and maintaining physical data centers and hard drives, you rent access to these technologies from providers like Amazon, Microsoft or Google.

The main Cloud Service Models

(i) IaaS (Infrastructure as a Service)

→ IaaS is the most flexible category that provides the fundamental building blocks of computing: virtual servers, storage and networking. You are responsible for managing the operating system, apps, and data, while the provider manages the physical hardware. Example product is Amazon Web Service (AWS) EC2.

(ii) PaaS (Platform as a Service)

* PaaS - Provide a framework that allows developers to build, test and deploy applications without worrying about the underlying infrastructure (Server/Storage)
* It simplifies the development process by handling the back "End Set Up Automatically".

Example: Google App Engine or Heroku.

(iii) SaaS (Software as a Service)

This known to be the most common model that delivers a complete software application over the internet, usually through a web browser.

The provider manages everything, from the hardware and servers to the code and security.

Example: Microsoft 365, Salesforce or Slack.

26. Version Control System (VCS)

* Is a Catagory of software tools that helps individuals and teams track changes to code or data over time. It act like a "Save" button with a memory, allowing you to record exactly who changed it, and why.
* Without Version control, teams often end up with confusing files with different versions.

Git

Git is the most popular distributed Version Control System in the world. Being distributed means that every developer has a full copy of the project history on their own computer, making it incredibly fast and reliable.

Data Analysts and Developers use it Because:

* Collaboration - Multiple people can work on the same script ~~start~~ simultaneously without over writing each other's work.
* If a new data model breaks or produces errors, you can instantly revert the entire project to a previous known good state.

It ability to prevent data loss and work seamlessly with platforms like GitHub.

Commit

A Commit is a Snapshot of your project at a Specific point in time. When you Commit your work, you are telling ~~the~~ Git to record the current State of all your files.

Branch

- A branch is an Independant line of development. The Main or Master branch usually Contains the Stable, finished version of your work.

27 Jupyter Notebook.

A Jupyter Notebook is an interactive environment or open-source web application that allows you to Create and Share documents containing live Codes equations, visualizations, and narrative text.

Programming Language.

The primary language used is Python:

Why Jupyter notebook popular?

* Interactive and step-by-step Execution. Unlike traditional programming where you have to run an entire script at once, Jupyter allows you to break your Codes into Cells. You can run one cell, see the results, and then move to the next.

28 Python

* Python is a high-level, interpreted programming language known for its clear Synthax and readibility. It was designed to be easy to write and even easier to Understand, Often resembling plain English.
* Python has gain its popularity over data analytics and AI due to easy to learn and use. It's clean, readable Synthax, requires few lines of Code Compared to many languages.
- Python has massive collection of Libraries Such as pre-written Codes Specifically designed for math, Statistics, and machine learning; so you don't have to build Everything from Scratch.

Four Python Libraries

(i) Numpy (Numerical Python) is the foundation for almost all math-heavy libraries. It allows Python perform Complex mathematical operations on large array and matrices extremely quickly.

(ii) Pandas → is the most important library for data manipulation. It introduces a structure called Dataframe, which is essentially a powerful, programmatic version of an Excel Spreadsheet.

(iii) Matplotlib | Seaborn → These are the primary libraries for data visualization. While Matplotlib provides the basic building blocks for any chart, Seaborn sits on top of it to make those charts look professional and publication-ready with very little code.

(iv) Scikit-learn → is the go to library for traditional Machine Learning. It contains simple efficient tools for predictive data analysis, covering everything from linear regression to complex clustering algorithms.

Section D: Data Ethics, Governance & Career:

29. Data Privacy

Data Privacy is a branch of data management that deals with the proper handling, processing, and storage of personal information. It focuses on how an individual's data should be collected, used, and protected according to legal requirements and ethical standards.

Importance of data privacy

* It ensures that people have a say in how their personal lives are digitized and tracked.
* Mismanaged data can lead to identity theft, financial fraud and stalking.
* Privacy prevents companies or governments from using personal data to unfairly discriminate against individuals.

The GDPR (General Data Protection Regulation.

* The GDPR – is the world's strongest and most comprehensive data privacy law. It went into effect on May 25, 2018, and was designed to modernize laws that protect the personal data of individuals in a digital age where data is constantly moved accross the borders.
* The GDPR primarily applies to the European Union and the European Economics Area, which include all 27 EU member states plus Iceland, Liechtenstein, and Norway.
* However the law applies to any organisation in the world, regardless where it is based if: It offers goods or service to people in the EU, Monitors the behaviour of people in the EU.

30 Data Ethics

* Data Ethics is a branch of ethics that evaluates the moral problems related to data, algorithms, and corresponding practices.
* It focuses on ensuring that data practice do not harm individuals or society, even if those practice are technically legal or profitable.

Three Examples of Unethical Use of Data or AI

① Biased AI in Hiring Systems → An AI system is trained on historical hiring data that favors certain groups of people.

* This unfairly excludes qualified candidates.

② Misuse of personal Data.

* The Facebook-Cambridge Analytical data Scandal – personal data from millions of users was collected without proper consent.

* This violates user privacy, and manipulates public opinion. Undermines trust in digital platforms.

③ Misinformation

* AI is used to create realistic fake videos or audio of people.

* This spreads false information and it damages reputations ~~Eff~~ especially of prominent people.

3. Data Governance

* is a collection of processes, roles, policies, standards, and metrics that ensures the effective and efficient use of information in enabling an organisation to achieve its goals.

- Data Governance is the strategic rules that defines who can take what action, upon what data, in what situation and using what method.

* Good Governance framework.

→ Data quality Management: It ensures data is accurate, complete and consistent

→ Data Policies and Standards → It define the rules for how data is collected, stored and used.

- Data Ownership and Stewardship – Ensures someone is responsible for data intergrity by assigning responsibilities to data owner and Data Steward.

32 <u>Bia in AI</u>

- Bias in AI Occurs when an AI System produces results that are systematically ~~Prej~~ Prejudiced, unfair, or Skewed in favor of or against certain groups of people or outcomes.

- An AI Model is Considered biased when:

It learns patterns that reflects real world inequalities

It Makes Consistantly inaccurate or unfair predictions for certain groups.

This is caused by :

Biased or Unrepresentative training data, incorrect labeling and design Choice that favours certain outcomes.

<u>Examples:</u>

Some Studies have shown that some facial recognition systems:

- perform better on lighter-skinned faces than on darker-skinned faces.

33.

Role	Main focus	key responsibility	Skills	common tools.
Data Analyst	Understanding data	- Analyze datasets - Create reports & dashboards - Identify trends	- SQL - Data visualization - Basic Statistics	Microsoft Excel, PowerBI Tableau
Data Scientist	Predictive modelling	- Build ML Models - Data Exploration - Feature Engineering	- Python / R - Statistics - Machine learning	- Jupyter notebook . Scikit learn.
Data Engineer	Data Infrastructure	- Build data pipelines - Manage datasets - Ensure data availability	- SQL - ETL Processes - Cloud Systems	- Apache Spark - Apache Airflow.
AI Engineer	AI System deployment	- deploy ML/AI Models - Optimize performance - Intergrate AI into APPS.	- Software Engineering - Deep Learning - APIs	TensorFlow, PyTorch.

34. Business Intelligence.

- Is a technology-driven process for analyzing data and delivering actionable information that helps executives, managers, and workers make informed business decision. BI turns raw data into meaningful insights to make better decisions.

Two Popular BI Tools.

① Power BI → Create interactive dashboards and reports. Therefore a Company can track monthly sales and instantly identities declining regions.

② Tableau → Transforms complex data into visual dashboards. Visualizing Customer behaviour to improve marketing strategies.

Section E: Personal Reflection.

35. One of the most interesting concepts in this Assignment is Generative AI and Computer vision.

This because of how it shifted from analyzing data to actually creating new content whether that's text, images, codes, or even music.

The fact that machines don't actually see like humans do; but they process images as numerical data. An image is converted into grid of pixel, each pixels has a numerical value (e.g RGB value for colour)

Somehow that has been stuck in my mind for a while, I don't know why but it make me curious about alot of thing I want to learn.

36. The concept that was most difficult to grasp is how neural networks actually learn. This because it involves multiple layers of computation happening simultaneously.

The resource that helped me understand is youtube and online blogs.

37. As someone who's working in the lab - I think AI is going to remove drudge work of data entry and repetive testing. This will allows me to spend more time on innovation, strategy, and solving the complex. Material challenges that a machine can't understand without my expertise.

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